IT2140- Database Design and Development

Year 2, Semester 1

Lab Sheet 02: SQL Table Joining

Objective

At the end of this lab session, students should be able to join tables and retrieve meaningful information by using INNER JOIN with related columns from multiple tables.

What is Table Joining?

When retrieving data from two or more related tables in a relational database, matching rows are **joined together** to produce the desired result. Rows are combined based on **related columns** (usually a foreign key referencing a primary key).

Example scenario:

If we want to list the names of employees along with their department names, we must combine data from both the emp and dept tables using the foreign key relationship.

Example 1

Question: Display the names of all employees along with the names of the departments they are assigned to.

Employee table

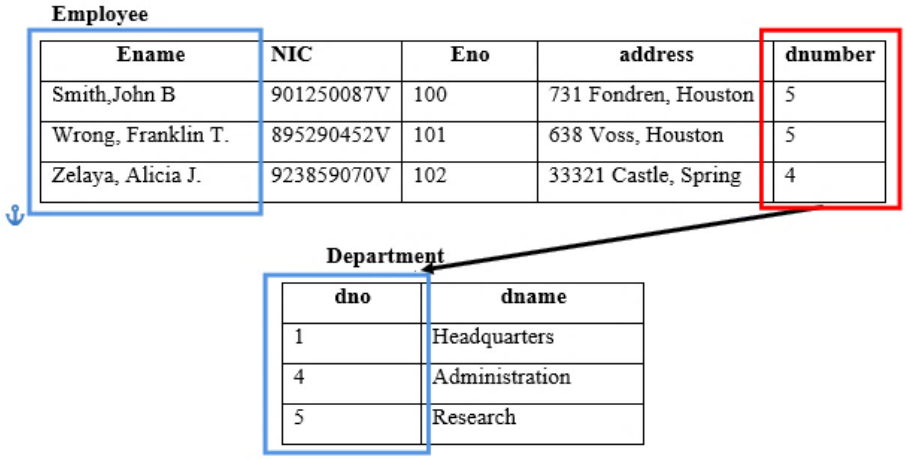
Ename	NIC	Eno	address	dnumber
Smith,John B	901250087V	100	731 Fondren, Houston	5
Wrong, Franklin T.	895290452V	101	638 Voss, Houston	5
Zelaya, Alicia J.	923859070V	102	33321 Castle, Spring	4

Department table

dno	dname
1	Headquarters
4	Administration
5	Research

Step 01: Examine the Tables

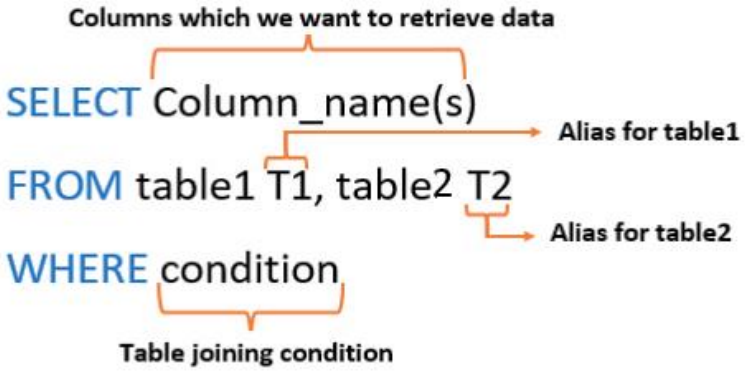
Before writing the query, first review the data in both tables to identify how they are related. Look for common columns (such as IDs) that can be used to join the tables correctly.



When examining the two tables, you'll notice that both contain department number columns (dnumber in the Employee table and dno in the Department table). The dnumber column in the Employee table refers to the dno in the Department table. Therefore, we can join these two tables using this common column to link employees with their respective departments.

Step 02: Construct the SQL Query

Now that the relationship between the tables is identified, let's write an SQL query to retrieve the required information by joining the related columns.



Alias

An **alias** provides a temporary name for a table or a column, making SQL queries shorter and easier to read. It only exists during the execution of the query.

Now, let's apply aliases and write the SQL query to retrieve the desired result.

```
SELECT E.Ename, D. dname
FROM Employee E, Department D
WHERE E. dnumber = D. dno
```

In this query, 'E' and 'D' are used as aliases for the Employee and Department tables, respectively. It is important to specify the **join condition accurately**; otherwise, the query may return incorrect or unintended results.

Step 03: View the Resulting Join

After executing the query, the two tables will be joined based on the specified condition. The structure of the combined table is shown in **Figure 1**, and the output of the query is displayed in **Figure 2**.

Ename	NIC	Eno	address	dnumber	dno	dname
Smith,John B	901250087V	100	731 Fondren, Houston	5	5	Research
Wrong, FranklinT.	895290452V	101	638 Voss, Houston	5	5	Research
Zelaya, Alicia J.	923859070V	102	33321 Castle, Spring	4	4	Administration

Figure 1:Joined Table

Ename	dname
Smith,John B	Research
Wrong, FranklinT.	Research
Zelaya, Alicia J.	Administration

Figure 2: Output Table

Example 2

Question: How many employees are there in each department? List the department name along with the number of employees.

Step 01: Examine the Tables

To retrieve the required information, we need to use both the Employee and Department tables, and apply the COUNT function to determine the number of employees in each department.

Step 02: Construct the SQL Query

```
SELECT D.dname, COUNT (*) AS 'No of employees'
FROM Employee E, Department D
WHERE E. dnumber = D. dno
```

Step 03: View the Resulting table

dname	No of Employees
Research	2
Administration	1